Universal rational parametrization keeps an integral cost

$$A_0 = t^{12}s^{28} \qquad B_0 = t^{7}s^{17} \qquad C_0 = t^{5}s^{12}$$
$$s = 1-t, \quad 0 < t < 1$$

$$A_0^{3} + B_0^{5} = C_0^{7}$$

$$A = \lambda^{35}A_0 \qquad B = \lambda^{21}B_0 \qquad C = \lambda^{15}C_0$$

The same positive rational λ must make all three bases primitive integers.